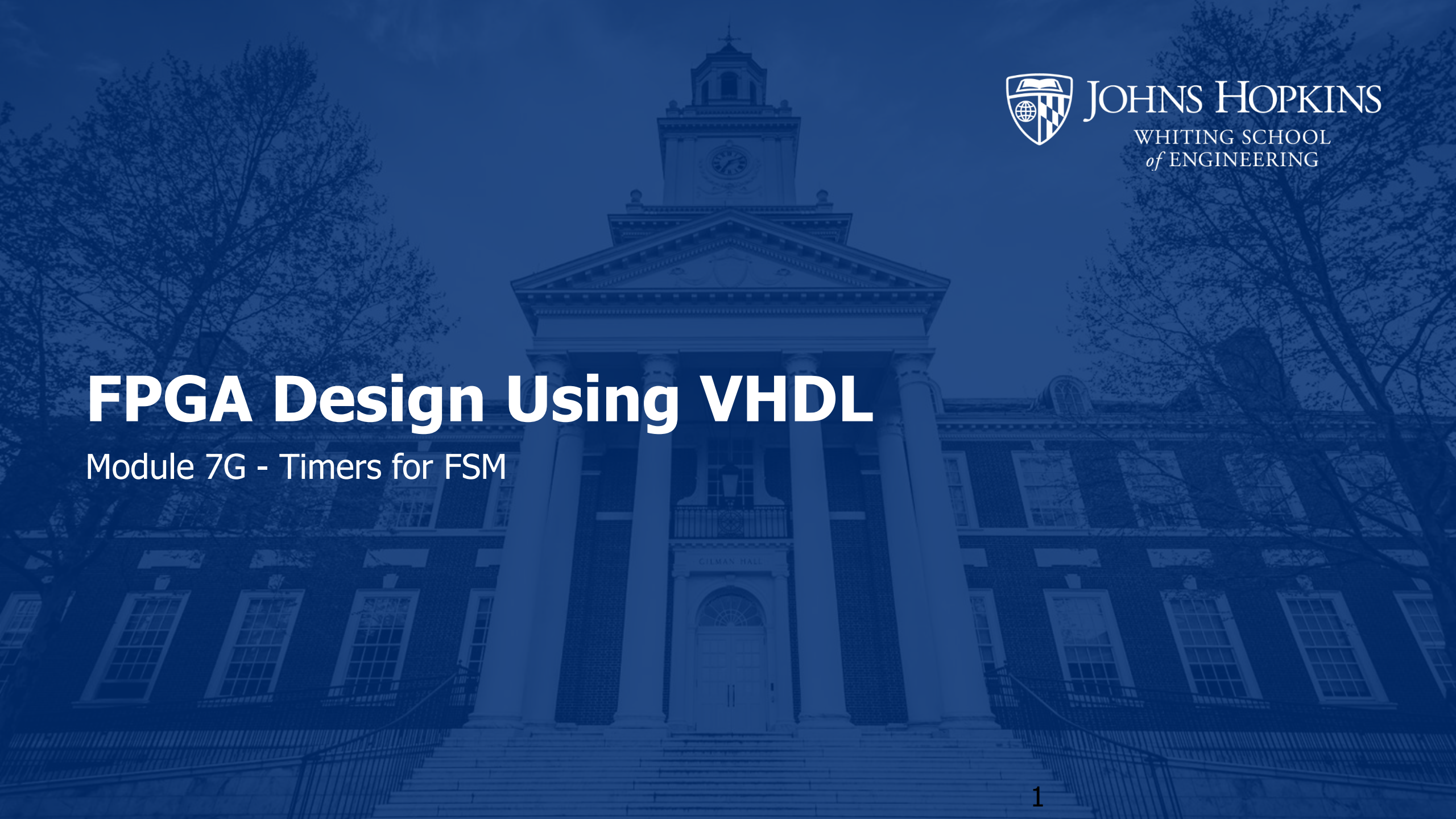
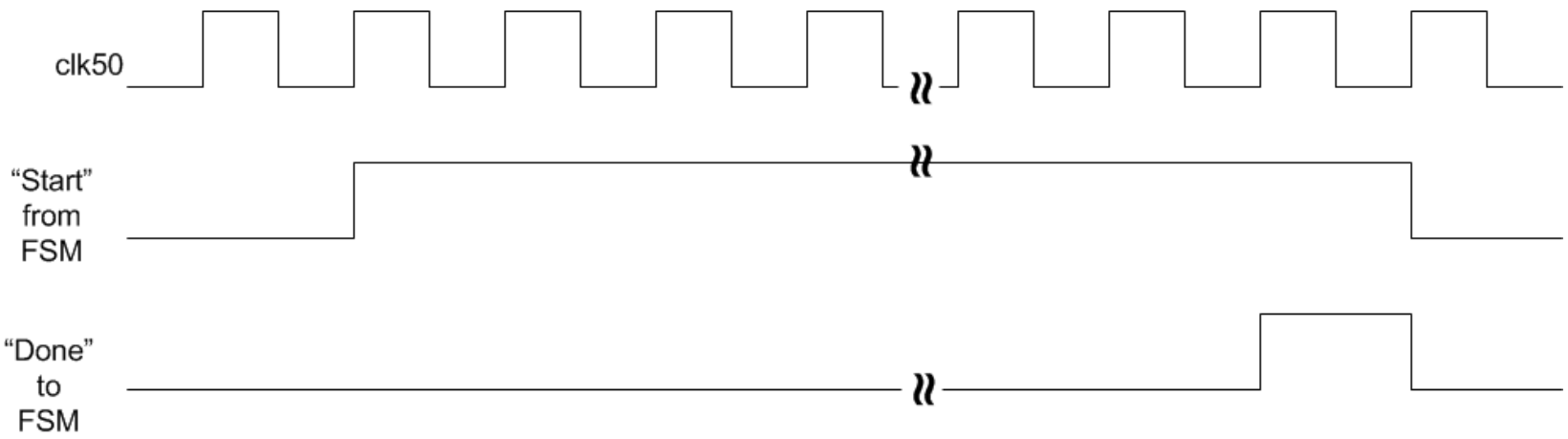


FPGA Design Using VHDL

Module 7G - Timers for FSM

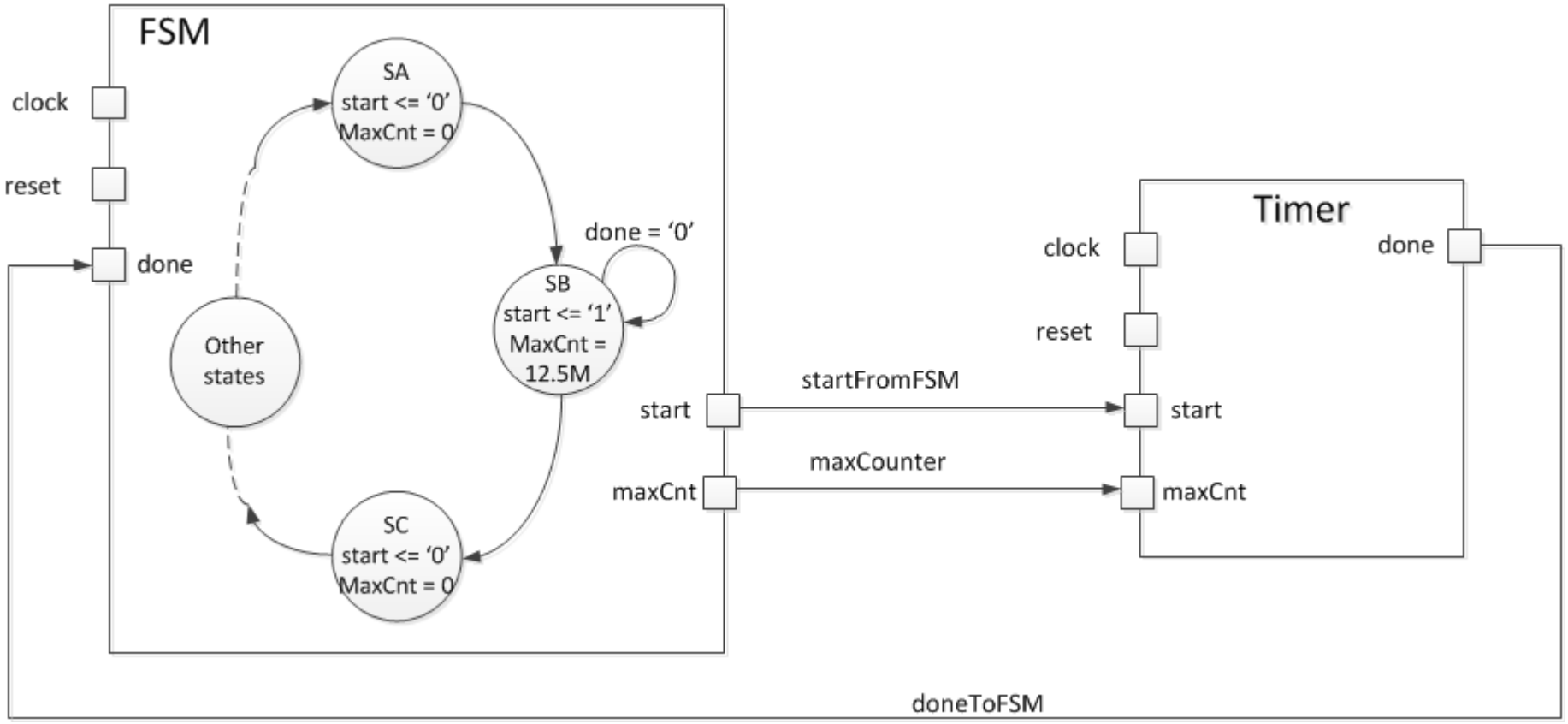
Timer Design

- Timer process that generates a “done” pulse upon receiving a “start” signal from one of the states in a FSM



- Remain in

Timer Design Continued



Timer Design – VHDL Code

- Feel free to use the following code or maybe you have a better timer

```
process(reset,clk)
begin
  if(reset = '1') then
     cntr <= (others => '0');
  elsif(rising_edge(clk)) then
     if(start = '1') then
          if(cntr < MaxCnt) then
               cntr <= cntr + 1;
          else
               cntr <= (others => '0');
          end if;
     else
          cntr <= (others => '0');
     end if;
  end if;
end process;
done <= '1' when cntr = MaxCnt else '0';
```